

FUNDAMENTAL IN AI AND ML PROJECT REPORT

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BOOK RECOMMENDATION SYSTEM –

Introduction

The Book Recommendation System is a Machine Learning based project that suggests books to users based on their interests. It uses content-based filtering techniques to recommend similar books by analyzing their genres and features.

Objective

- To build a system that recommends books based on user input
- To understand basic Machine Learning concepts
- To implement similarity-based recommendation using NLP techniques.

Technologies Used

- Python
- Pandas
- Scikit-learn
- VS Code

Methodology

The project follows these steps:

1. Create a dataset containing book titles and genres
2. Convert text data into numerical form using CountVectorizer
3. Calculate similarity between books using cosine similarity
4. Recommend top similar books based on user input

Algorithm Used

Content-Based Filtering

This method recommends items similar to what the user likes based on item features.

Cosine Similarity

It measures similarity between two vectors:

$$\text{Cosine Similarity} = (A \cdot B) / (||A|| ||B||)$$

Implementation

Dataset is created manually within the code

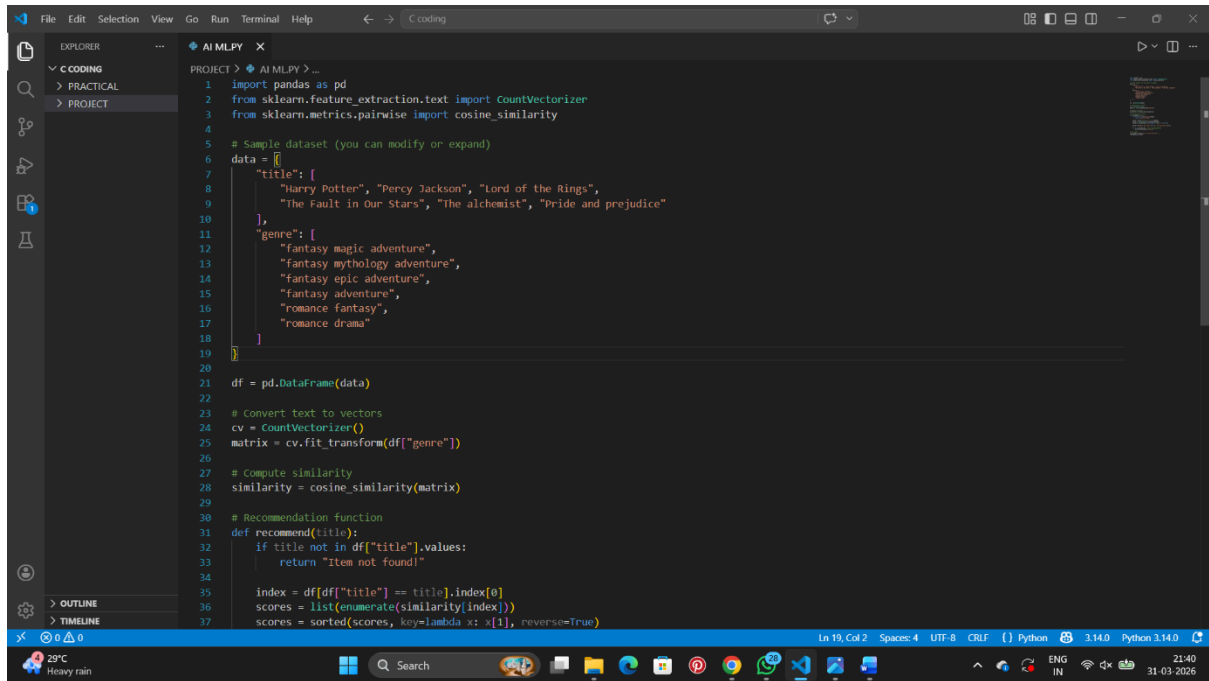
Text features are extracted using
CountVectorizer

Similarity matrix is computed using cosine
similarity

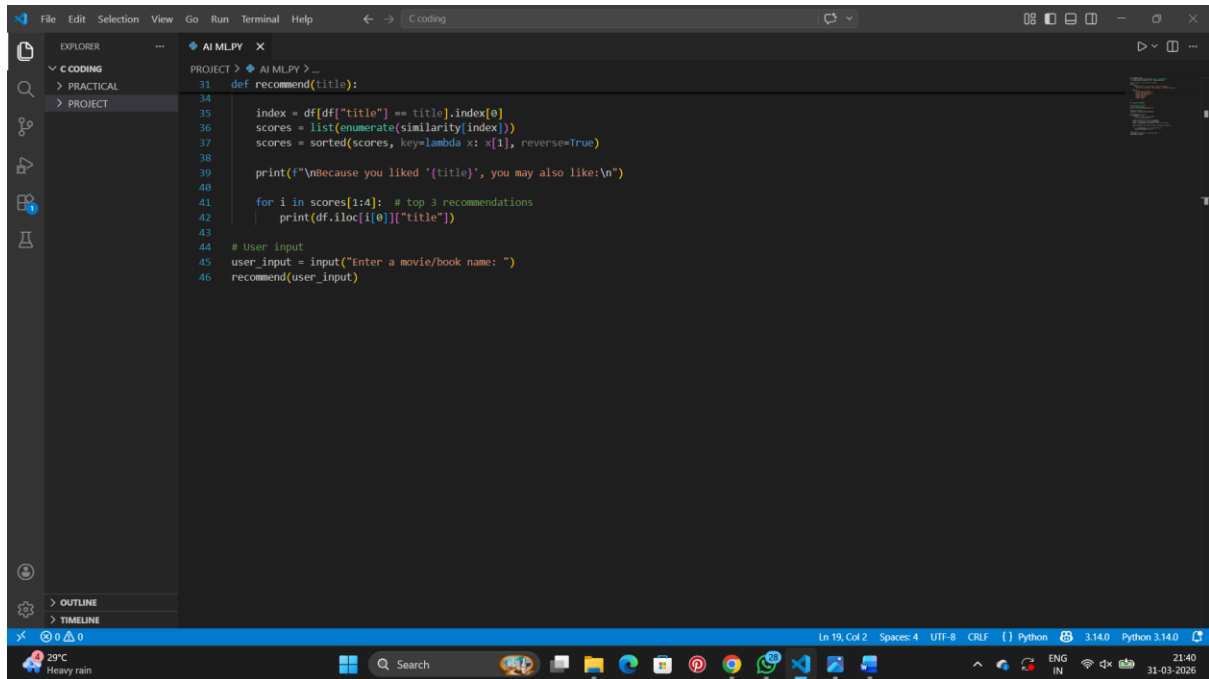
User inputs a book name

System returns top 3 similar books

Sample Code



```
1 import pandas as pd
2 from sklearn.feature_extraction.text import CountVectorizer
3 from sklearn.metrics.pairwise import cosine_similarity
4
5 # Sample dataset (you can modify or expand)
6 data = {
7     "title": [
8         "Harry Potter", "Percy Jackson", "Lord of the Rings",
9         "The Fault in Our Stars", "The alchemist", "Pride and prejudice"
10    ],
11    "genre": [
12        "fantasy magic adventure",
13        "fantasy mythology adventure",
14        "fantasy epic adventure",
15        "fantasy adventure",
16        "romance fantasy",
17        "romance drama"
18    ]
19 }
20
21 df = pd.DataFrame(data)
22
23 # Convert text to vectors
24 cv = CountVectorizer()
25 matrix = cv.fit_transform(df["genre"])
26
27 # Compute similarity
28 similarity = cosine_similarity(matrix)
29
30 # Recommendation function
31 def recommend(title):
32     if title not in df["title"].values:
33         return "Item not found!"
34
35     index = df[df["title"] == title].index[0]
36     scores = list(enumerate(similarity[index]))
37     scores = sorted(scores, key=lambda x: x[1], reverse=True)
```



```
31 def recommend(title):
32
33     index = df[df["title"] == title].index[0]
34     scores = list(enumerate(similarity[index]))
35     scores = sorted(scores, key=lambda x: x[1], reverse=True)
36
37     print(f"\nBecause you liked '{title}', you may also like:\n")
38
39     for i in scores[1:4]: # top 3 recommendations
40         print(df.iloc[i][0]["title"])
41
42 # User input
43 user_input = input("Enter a movie/book name: ")
44 recommend(user_input)
```